



University of Colombo, Sri Lanka



University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
BACHELOR OF SCIENCE HONOURS IN COMPUTER SCIENCE
BACHELOR OF SCIENCE HONOURS IN SOFTWARE ENGINEERING
BACHELOR OF SCIENCE IN INFORMATION SYSTEMS
BACHELOR OF SCIENCE HONOURS IN INFORMATION SYSTEMS



Third Year Examination - Semester I - UCSC AY21[held in September/October 2024]

SCS3203/IS3108 — Middleware Architecture

(Two (2) Hours)

Answer ALL questions

Number of Pages = 10

Number of Questions = 4

To be completed by the candidate

Index Number:

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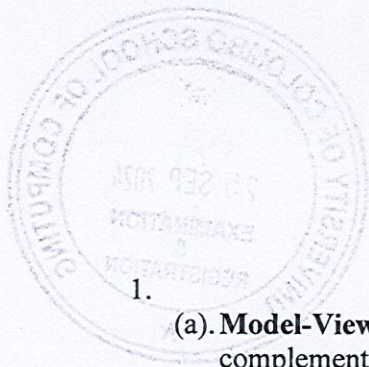
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Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 10 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smart watches, and mobile phones, are **not allowed**.
- VII. Do not tear off any part of this answer book. Under no circumstances may a candidate remove this book, used or unused, from the Examination Hall.

To be completed by the examiners

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1.

- (a). **Model-View-Controller (MVC)** and **3-Tier** are two popular architectures that complement each other in most industry implementations. Explain how these two architectures complement each other in building a thin-client, multi-tiered system, and discuss how both can coexist in such a system to provide a robust and scalable solution.

[6 Marks]

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- (b). Discuss **three (3)** differences between **Service-Oriented Architecture (SOA)** and **MicroServices Architecture (MSA)** in terms of service composition, data persistence and communication among services.

[6 Marks]

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- (c). In a 2-tier client-server architecture, the client and server typically communicate using a request-response model, which follows a synchronous messaging pattern. To transition from this synchronous model to an asynchronous model, resulting in a request-async response pattern, additional improvements are needed. Describe the enhancements required to convert the synchronous request-response communication between a client and server into an asynchronous request-async response model.

[7 Marks]

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- (d). **Reliability** and **Availability** are two important indicators of the nature of fault tolerance in a platform. Mean Time To Fail (MTTF) and Mean Time To Repair (MTTR) are popular measurements for calculating the Reliability and Availability of platforms. Consider the following two platforms:

Platform 1: MTTF is 6 months and MTTR is 24 hours

Platform 2: MTTF is 2 months and MTTR is 1 hour.

Discuss the relative differences in the availability and reliability of these two platforms.

[6 Marks]

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2.

(a). A **Digital Internet and Mobile Banking platform** uses **Microservices Architecture (MSA)** to implement the core business services. Among these services, the **customer profile identification service** is highly critical, as many other services depend on it. . As a result, the availability and reliability of this service are important to the overall performance and functionality of the platform.

- (i). Describe **three (3)** strategies to handle potential failures of the **customer profile identification service** to make sure the overall platform adapts and recovers to maintain normal functionality.

[6 Marks]

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- (ii). In the **Microservices Architecture (MSA)**, there are several **one-to-one** client-service interaction styles based on **synchronous** and **asynchronous** dimensions. Propose and justify the most suitable client service interaction style to interact with the **customer profile identification service**, especially in situations where the request throughput increases rapidly.

[8 Marks]

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- (b). **API Gateway** is an important component of the **Microservices Architecture (MSA)**. Describe **four (4)** key responsibilities of an API gateway in this architecture. Additionally, discuss **one (1)** potential drawback associated with the use of an API Gateway..

[5 Marks]

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- (c). Microservices Architecture (MSA) promotes **polyglot persistence** as opposed to a **centralized enterprise data model** approach that is established in the industry. Polyglot persistence together with **event-driven service architecture** enables innovative models for realising service interaction styles. Explain **three (3)** challenges an architect might face when implementing services based on a polyglot event-driven model within a Microservices Architecture framework..

[6 Marks]

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3.

- (a). A popular apparel-producing enterprise in Sri Lanka called “Raggedy Ralph” (RR), has implemented a **distributed services architecture** for the **merchandising** departments to automate the quotation generation process for customers. One of the major responsibilities of a merchandising department is to generate **proposals/quotations** for customer orders. RR has many subsidiaries producing different apparel products such as Jeans, Sportswear and Shirts. Each subsidiary specialises in a specialised product – for example, the “Raggedy Ralph-Jeans” subsidiary specialises in producing ladies’ and men’s Jeans. The central merchandising department of RR receives orders from customers and each order may contain items from different product lines – for example, an order may contain 100K items of Jeans, 200K items of shirts, etc. The central merchandising service needs to break down the customer order into specific product types and request quotations from each respective subsidiary responsible for each product type. The responses from all subsidiaries for a given order are amalgamated by the central merchandising exchange to be presented to the customer as the final proposal/quotation.
- (i). Propose an **architectural pattern** that is most suitable for implementing the distributed services model for the above-mentioned scenario. Provide a **justification** for your choice of architectural pattern.

[4 Marks]

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- (ii). The proposed architectural pattern may require several **integration patterns** to realise the final quotation from the customer order consisting of different product categories. Identify and describe at least **three (3)** integration patterns that would work together to achieve the overall function of formulating a proposal as a reply to a customer order request. Additionally, illustrate the connectivity of these integration patterns in a diagram.

[8 Marks]

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- (b). The **Enterprise Service Bus (ESB)**, **Publish-Subscribe Message Channels (Pub-Sub)** and **Object Request Broker (ORB)** represent the middleware components from several popular architectural patterns. Explain **one (1)** unique feature in each of the three (3) middleware components in comparison to the others.

[6 Marks]

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- (c). **Remote Method Invocation (RMI)** and **Common Object Request Broker Architecture (CORBA)** are popular distributed services architecture implementations. Explain **two (2)** similarities and **two (2)** differences between RMI and CORBA architectures.

[7 Marks]

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4.

- (a). **Virtualisation** uses a software component called the **hypervisor** to create **Virtual Machines (VMs)** which act as separate servers running on their own Operating Systems. However, the recent technologies related to **Containerisation** offer several advantages over Virtualisation technology. Explain **three (3)** advantages of using Containerisation over Virtualisation in the context of deploying services within the Microservices Architecture implementations.

[6 Marks]

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- (b). **Docker** is the most widespread containerised technology in the industry today. **Kubernetes** is another complementary technology used with Docker containerisation in the industry. Consider a scenario where a Microservices Architecture consisting of over 1000 services are implemented using the Docker container technology. Explain the primary role of Kubernetes in managing such a containerized environment. Discuss whether Kubernetes is necessary for this implementation and provide justification for your response.

[7 Marks]

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- (c). **Concurrent transactions** may access shared objects/data to complete a sequence of atomic steps of a transaction. **Two-phase locking (2-PL)** is a popular method for securely updating shared data during concurrent transactions. However, 2-PL does not guarantee the completion of a transaction and may result in **restricted situations**. Explain a popular restrictive situation that may occur during concurrent transactions and propose a method to resolve or overcome such a situation.

[6 Marks]

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- (d). The business logic tier of a popular e-commerce platform was implemented about 10 years ago based on the **Service Oriented Architecture (SOA)**. Based on the recent advances in **Microservices Architecture (MSA)** together with Containerised technology, the principal architect has decided to migrate the SOA-based architecture to a fully distributed MSA and containerised platform. Explain **three (3)** strategies that could be adopted by the principal architect to migrate the SOA-based architecture to MSA.

[6 Marks]

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